

The sound-of-words model: A developmental perspective of phonolexical acquisition

Thierry Nazzi*

Language and Cognition Team, Integrative Neuroscience and Cognition Center, CNRS- Université Paris Cité, Paris, France

*Corresponding authors: e-mail address: thierry.nazzi@parisdescartes.fr.

Contents

1. The "division of labor" hypothesis	136
2. Evaluating the C-bias in lexical processing in adulthood	138
3. Hypotheses regarding the origin of the C-bias	141
4. Evaluating the C-bias in lexical processing in infancy and toddlerhood	143
5. Romance languages	144
6. Germanic languages	146
7. Sinitic languages	147
8. Semitic languages	148
9. The sound-of-words model	149
10. Concluding remarks	153
Acknowledgments	153
References	154

Abstract

The current chapter reviews 25 years of research on the so-called consonant bias in lexical processing, according to which consonants, rather than vowels, are most relevant to build the lexicon. The evidence so far suggests the C-bias might be prevalent in adulthood, though more work is needed on tone languages that might change this view. The findings from developmental studies offer a more nuanced approach, showing important crosslinguistic differences in the timing of acquisition of asymmetrical processing of consonants and vowels in lexically-related processes, and in the direction of the bias observed. This suggests that the bias is acquired on the basis of phonological and lexical cues. We propose a new model for the acquisition of these biases, the Sound-of-Words model, that could be used to guide further research on this issue.

Keywords: C-bias, Crosslinguistic variation, Early language acquisition, Infants, Language acquisition, Lexical acquisition, Lexicon, Phonological acquisition, Phonology

Language acquisition is a crucial aspect of human development, being the single best predictor of school achievements in all subjects in primary school (third and fifth grades; [Pace et al., 2018](#)). Language is a complex construct, which can be described at different levels organized around fundamental building blocks, such as: the phonological level around phonemes and prosodic units; the lexical level around words and morphemes; the syntactic level around grammatical categories and rules. This general organization is shared by all human languages. At the same time, languages differ in crucial ways on all these levels. The existence of these commonalities and variabilities across languages has consequences for language processing and acquisition, and the aim of developmental psycholinguistics is to determine the mechanisms that allow infants to learn the different linguistic levels of their native language(s). This raises several central theoretical issues for infant development, such as the contribution of nature and nurture to acquisition, the links between the acquisition of the different linguistic levels, and the cascading developmental effects and predictive effects of early acquisitions on later developmental outcomes. The current chapter explores how the investigation of the emergence of processing biases in the acquisition of the phono-lexicon, namely the interface between the phonological system and the lexicon, brings new, related knowledge on these issues, and we propose a new model of the emergence of this interface, the Sound-of-Words model. According to this model, the emergence of a Sound-of-Words bias constitutes a major developmental milestone, attesting the binding of the phonological and lexical levels, and bootstrapping subsequent vocabulary development.



1. The “division of labor” hypothesis

This research takes as its premise the “division of labor” hypothesis ([Nespor et al., 2003](#)) by which two complementary phonological biases impact language processing and acquisition: “[C]onsonants, rather than vowels, are most relevant to build the lexicon, and vowels, rather than consonants, are most relevant for grammatical [and prosodic] information.” This proposal was based on several linguistic observations, and emerging experimental data suggesting different contributions of consonants and vowels to lexical access in adulthood. For example, it underlined the fact that consonants are more numerous than vowels in most languages ([Ladefoged & Maddieson, 1996](#)),

that lexical roots are mostly made up of consonants in Semitic languages, or that consonants tend to disharmonize within words (contrary to vowels). All these factors would make consonants more informative than vowels at the lexical level.

At the processing level, there was evidence that consonants and vowels tend to be processed differently, with more categorical perception of consonants than vowels in both adults (Fry et al., 1962; Liberman et al., 1957) and infants (Eimas et al., 1971; Kuhl et al., 1992). There is also evidence for differences in the timing of acquisition of consonants and vowels, with an earlier acquisition of native vowels – starting by 6 months– compared to native consonants –starting by 10 months (e.g., Best et al., 1988; Kuhl et al., 1992; Werker & Tees, 1984). Neuropsychological evidence also described two aphasic patients who presented contrasted error patterns when producing consonants and vowels, suggesting the involvement of different brain regions in the processing of the two types of phonemes (Caramazza et al., 2000). Moreover, the first experimental evidence in support of the division-of-labor proposal, revealing a consonant bias in lexical processing, was a series of word reconstruction studies conducted on Dutch-, English- and Spanish-speaking adults (Cutler et al., 2000; van Ooijen, 1996–). In these studies, adults were presented with pseudowords and asked to change a phoneme of each pseudoword to make it a word. Pseudowords (e.g. /kebra/) could be modified by changing either a consonant (/zebra/) or a vowel (/cobra/). In all three languages, they observed that adults retrieved words by changing more often and more rapidly a vowel into another vowel than a consonant into another consonant, establishing that consonants are stronger cues to word identity.

The importance of this “division of labor” proposal lies in the fact that if these two processing biases are present early in life, then they might bootstrap lexical acquisition, helping infants learn the words of their native language on the one hand, its grammatical and prosodic properties on the other hand. In the following, we evaluate the evidence that has been gathered since this proposal was made, first reviewing adult data, then developmental data. Since little work has been done on the grammatical/prosodic side of the “division of labor” proposal (see, however, Toro et al., 2008 for adult work; Hochmann et al., 2011, and Pons & Toro, 2010, for infant work), the present chapter focuses on lexical processing and the so-called “C-bias,” according to which it is easier to learn and retrieve words on the basis of their consonants than on the basis of their vowels.



2. Evaluating the C-bias in lexical processing in adulthood

Further adult studies (see review by [Nazzi & Cutler, 2019](#)), using speech perception tasks tapping into word learning and lexical access, tested whether the C-bias in lexical processing is present cross-linguistically as predicted if it is an innate, universal bias. Various tasks were used, three of them tapping into processes of accessing known words: word reconstruction (in which adults are presented with a pseudoword and have to transform it into a word by changing either a consonant or a vowel), word spotting (in which they have to spot a word embedded in a longer pseudoword), and primed lexical decision (in which they have to decide whether a target is a word, the target being preceded by a prime that is either unrelated, or related to the target by its consonants or its vowels). Two other tasks are related to the learning of new word forms or new words: word segmentation (in which adults hear a continuous speech stream made up of a few pseudowords, and have to use distributional properties to find the words that make up the language), and word learning (in which they are taught pairs of new words differing by a consonant or a vowel).

At present, studies have been conducted in 16 languages, most of them Indo-European languages: English, Dutch, German, Danish, Norwegian, French, Italian, Spanish, Slovak and Russian. A few languages from other linguistic families have also been investigated: Sesotho, Berber, Japanese, Mandarin, Cantonese and Hebrew. A summary of the results found is presented in [Table 1](#). For the Indo-European languages, on which most studies were conducted, the evidence overwhelmingly establishes a C-bias in all of these lexically-related processes, with just a few studies failing to show a bias (note that there is only one study on both Danish and Norwegian, failing to provide conclusive evidence). A C-bias was also found in Sesotho, Japanese and Hebrew, suggesting this bias extends beyond Indo-European languages, although no evidence was found in Berber. In contrast, results from Mandarin and Cantonese, two tone languages, fail to show evidence of a C-bias, and provide emerging evidence that a V-bias might be present in some languages ([Gómez et al., 2018](#); [Poltock et al., 2018](#); [Wiener, 2020](#); [Wiener & Turnbull, 2016](#)).

Hence, if a C-bias has largely been found, which would support the idea of a language-general C-bias in lexical processing, two limits have to be pointed out. First, these findings mostly come from Indo-European languages, with only a few of the around 7000 other languages investigated.

Table 1 Crosslinguistic consistency in adult auditory lexical processing (C: consonant bias; V: vowel bias; V + T: advantage of tone-carrying vowel; C-V n.d., no difference).

Languages	Tasks	Lexical bias	Reference
English	Word reconstruction	C	Van Ooijen (1996); Sharp et al. (2005); Moates et al. (2002)
	Word spotting	C	Norris et al. (1997, 2001); Newman et al. (2011)
	Syllable detection	C-V n.d.	Kearns et al. (2002)
	Primed lexical decision	C	Delle Luche et al. (2014)
	Word segmentation	C-V n.d.	Newport & Aslin (2004)
	Word learning	C	Creel et al. (2006); Escudero et al. (2016)
Dutch	Word reconstruction	C	Cutler et al. (2000)
	Word spotting	C	McQueen et al. (1998)
	Primed lexical decision	C-V n.d.	Cutler et al. (1999)
German	Word spotting	C	Hanulíková et al. (2011)
Danish	Word segmentation	C-V n.d.	Trecca et al. (2019)
Norwegian	Word segmentation	C-V n.d.	Trecca et al. (2019)
Spanish	Word reconstruction	C	Cutler et al. (2000), Marks et al. (2002)
French	Word spotting	C	Dumay et al. (2002)
	Primed lexical decision	C	New et al. (2008); New & Nazzi (2014); Delle Luche et al. (2014); Schmandt et al. (2022)
	Word segmentation	C	Bonatti et al. (2005), Mehler et al. (2006)
	Word learning	C	Havy et al. (2014)

Continued

Table 1 Crosslinguistic consistency in adult auditory lexical processing (C: consonant bias; V: vowel bias; $V + T$: advantage of tone-carrying vowel; C-V n.d., no difference).—cont'd

Languages	Tasks	Lexical bias	Reference
Italian	Word segmentation	C	Toro et al. (2008)
Slovak	Word spotting	C	Hanulíková et al. (2011)
Russian	Word spotting	C-V n.d.	Alexeeva et al. (2017)
	Word segmentation	C	Gomez et al. (2018)
Sesotho	Word spotting	C	Cutler et al. (2002a)
Berber	Word spotting	C-V n.d.	El Aissati et al. (2012)
Japanese	Word spotting	C	McQueen et al. (2001) ; Cutler et al. (2009)
	Word reconstruction	C	Cutler and Otake (2002)
Hebrew	Primed lexical decision	C	Lador-Weizman and Deutsch (2022)
	Word segmentation	C	Lador-Weizman and Deutsch (2024)
Mandarin	Word reconstruction	V	Wiener and Turnbull (2016) ; Wiener (2020)
	Word segmentation	$V + T$	Gomez et al. (2018)
	Word learning (Cantonese)	C-V n.d.	Poltock et al. (2018)
Cantonese	Word segmentation	$V + T$	Gomez et al. (2018)
	Morpheme-spotting	C	Yip (2004a,b)
	Word learning	C-V n.d.	Poltock et al. (2018)

It is thus crucial to further extend this kind of research to a larger range of languages and language families, in order to further probe the validity of such language-general claim. Second, if a V-bias in lexical processing is a feature that extends beyond Mandarin and Cantonese to all or most tone languages, this would challenge the view that the C-bias is the predominant pattern in lexically-related processing, given that tone languages constitute the majority of the languages of the world. Further studies on tone languages are thus key to confirm or refute the predominance of the C-bias. While waiting for further cross-linguistic extensions (that would allow us to specify the factors that modulate the expression of the C-bias in a given language), let us review whether a C-bias is found in development, how it might be set into place, and whether it is modulated cross-linguistically.



3. Hypotheses regarding the origin of the C-bias

Within the framework of phonological bootstrapping (Morgan & Demuth, 1996), the C-bias in lexical processing, if functional early in development, is viewed as a powerful tool for infants to bootstrap into lexical and syntactic acquisition. In terms of the origins of the C-bias, different hypotheses have been considered in the literature, making different predictions about the developmental timing at which the C-bias would be functional, and the existence of crosslinguistic modulation in its expression. According to the first one, or initial bias hypothesis, the C-bias would be innate, “human beings com[ing] into the world knowing that languages are structured in such a way” (Nespor et al., 2003, p. 224; see also Bonatti et al., 2007; Pons & Toro, 2010). This predicts that infants would potentially process consonants and vowels as distinctive linguistic categories from birth. It further predicts that a C-bias in “lexically-related” tasks should be observed across languages, irrespective of the distributional/acoustic characteristics of each language. A weaker version of this position is that the C-bias should be the predominant bias found cross-linguistically.

In addition, the C-bias might also result from the properties of the linguistic input infants are exposed to, and thus constitute a case of early attunement to the native language(s). Floccia et al. (2014) discussed acoustic/phonetic and lexical factors potentially involved in the emergence of the C-bias, and three acquisition hypotheses can be found in the literature: the acoustic/phonetic bias hypothesis (Bouchon et al., 2015; Floccia et al., 2014), the lexical bias hypothesis (based on Keidel et al., 2007), and the

phono-lexical hypothesis (Nishibayashi & Nazzi, 2016; Poltrock & Nazzi, 2015).

According to the acoustic/phonetic bias hypothesis, the C-bias emerges as a result of differences in acoustic properties, as consonants tend to be shorter, less periodic and steady but tend to be perceived more categorically than vowels, leading to the construction of two distinct categories (Bouchon et al., 2015; Floccia et al., 2014). Since phonological acquisition starts during the first year of life (Best et al., 1988; Kuhl et al., 1992; Werker & Tees, 1984), the C-bias could emerge before the first birthday, independently of whether or not infants have acquired a sizeable lexicon. Moreover, it should be modulated over the course of development by the acoustic/phonetic characteristics of the ambient language, so that differences in the age at which the bias emerges and its strength should be found cross-linguistically.

According to the lexical bias hypothesis, the emergence of the C-bias would be related to lexical factors, linked to the phonological structure of the lexicon in acquisition. The C-bias would emerge as a result of the fact that, compared to vowels, consonants would be more informative cues to lexical identity in the native language. Work on English and French suggests that this higher informativeness of consonants could be related to density of phonological neighborhoods linked to C- and V-skeletons (Delle Luche et al., 2014; Keidel et al., 2007; New & Nazzi, 2014; Schmandt et al., 2022). Since calculating informativeness requires to have a sufficiently large lexicon, it was proposed that such a critical mass would not be reached before some point during the second year of life at the earliest, when lexical acquisition appears to accelerate. Hence, the lexical bias hypothesis predicts a later acquisition of the C-bias than the acoustic/phonetic bias hypothesis. Moreover, the lexical hypothesis also predicts crosslinguistic differences, since differences in the structure of lexicons across languages should impact the relative informativeness of consonants and vowels, and the ability and speed at which these properties can be acquired.

More recently, Poltrock and Nazzi (2015) proposed that rather than considering phonological and lexical properties as mutually exclusive sources from which the C-bias could be acquired, both should in fact contribute to the emergence of the C-bias. This proposal was based on adult work showing that both phonological and lexically-related properties contribute to lexical access in adulthood (New & Nazzi, 2014; Schmandt et al., 2022). It was also taking into account increasing evidence that lexical acquisition starts in the first year of life, as attested by studies showing that infants start recognizing the sound pattern of their name by 5 months

Table 2 Crosslinguistic and developmental predictions according to the different hypotheses explaining the origin of the C-bias.

Hypotheses	Origin	Cross-linguistic differences?	Timing of expression?
Initial	Innate	No	At birth
Acoustic/ phonetic	Learned through acoustic/ phonetic acquisition	Yes	Before 12 months
Lexical	Learned through experience with the native lexicon	Yes	After 12 months (sizeable lexicon)
Phono-lexical	Learned through phonological and lexical experience	Yes	Before 12 months

(Bouchon et al., 2015; Mandel et al., 1995) and of familiar words (Hallé & de Boysson-Bardies, 1994; Swingley, 2005; Vihman et al., 2004) by 11 months, and start comprehending some words by 6 months (Bergelson & Swingley, 2012; Tincoff & Jusczyk, 1999, 2012). It further considered findings establishing that phonological and lexical acquisitions are intertwined (Feldman et al., 2013; Stager & Werker, 1997; Yeung & Nazzi, 2014; Yeung & Werker, 2009). Lastly, it considered that acquisition of lexically-related properties could take place at the level of the pre-lexicon (including all word forms known by an infant) rather than the lexicon per se. Since infants' ability to segment word forms from fluent speech is found by 4 to 6 months in several languages (Berdasco-Muñoz et al., 2018; Bosch et al., 2013; Goyet et al., 2013; Höhle & Weissenborn, 2003; Jusczyk & Aslin, 1995; Kooijman et al., 2013), this pre-lexicon might become large enough to sustain the acquisition of lexically-related properties, and the emergence of the C-bias, before the end of the first year of life. As the other hypotheses stating that the C-bias is acquired, this phono-lexical hypothesis also predicts crosslinguistic differences in the emergence of the C-bias. Table 2 presents the different hypotheses proposed and their crosslinguistic and timing predictions.



4. Evaluating the C-bias in lexical processing in infancy and toddlerhood

To explore the innate and/or acquired determinants of the C-bias, a range of studies, starting from Nazzi (2005) with French-learning toddlers,

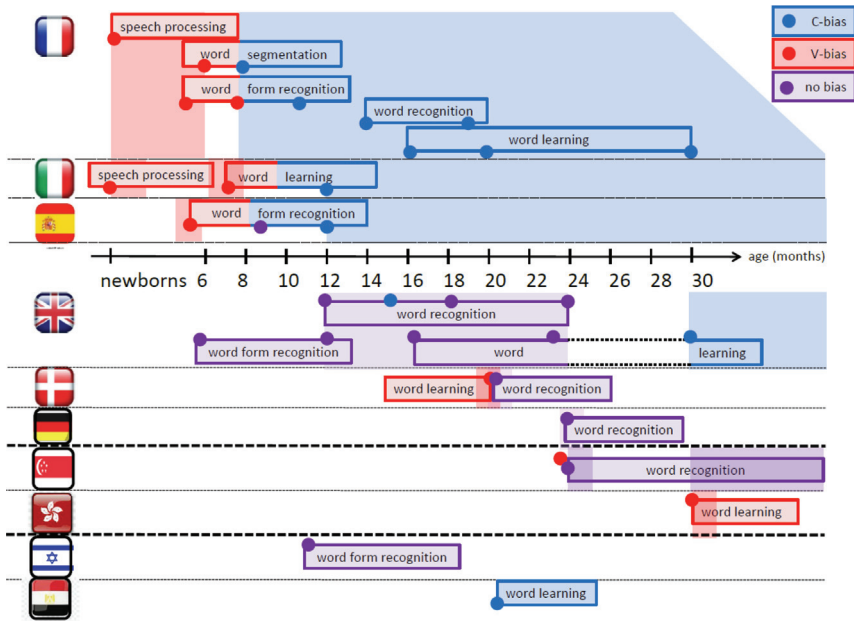


Fig. 1 Crosslinguistic data on C- and V-biases in infant/toddler lexically-related processing across languages. From top to bottom: Romance (French, Italian, Spanish), Germanic (English, Danish, German), Sinitic (Mandarin, Cantonese) and Semitic (Hebrew, Arabic) languages.

have been conducted in infancy and early childhood, in different linguistic populations. Most of these studies examined either infants' ability to learn pairs of new words that differ by one consonant or vowel, or their sensitivity to consonant or vowel mispronunciations of previously heard words or word forms. At this stage, the results found in languages from three different linguistic families suggest three different developmental trajectories (see review by [Nazzi & Cutler, 2019](#), and visual summary in [Fig. 1](#)).



5. Romance languages

The first developmental trajectory was found for three Romance languages, and corresponds to an early acquisition of a C-bias. In French, the C-bias was first established by [Nazzi \(2005\)](#), in which 20-month-old French-learning toddlers were presented with triads of novel objects, comprising two objects given the same name and a third object given a name that differed by one phonetic feature. Following this learning phase, they were asked to group

together the two objects with the same name. Infants successfully learned the object names when they differed by one of their consonants (e.g., /pize/ vs. /tize/) but not when they differed by one of their vowels (/pize/ vs. /pyze/). Further word-learning studies with French-learning toddlers established that a C-bias is found from 16 months to 6 years of age (Havy et al., 2011; Havy & Nazzi, 2009, 2014; Nazzi et al., 2009). A C-bias was also found in conflict studies in which, following a phase in which children learn the names of two objects (e.g., a /tide/ or a /pyde/), they were then asked to match a third name (e.g., a /pide/) to one of the preceding objects, which required neglecting either the consonantal or the vocalic mismatch. Although there is some variability in performance depending on the specific consonant and vowel contrasts tested, the C-bias in French appears to be a robust phenomenon. Indeed, it was found across different types of consonants (e.g., /nuk/ vs. /muk/, /rize/ vs. /lize/; Nazzi & New, 2007), different lexical structures (mono- and disyllables; e.g., Nazzi, 2005), and different positions within words (word-initial or not) and syllables (onset versus coda; e.g., Havy et al., 2014; Nazzi & Bertoncini, 2009; Nazzi & Polka, 2018; Nazzi et al., 2016). The C-bias also appears when toddlers are tested on their recognition of known words, for which they appear more sensitive to consonant than vowel mispronunciations (at 14 months: Zesiger & Jöhr, 2011; at 19 months: Von Holzen et al., 2023).

Additional studies on French explored whether the C-bias in lexically-related processes extends to the first year of life. To do so, infants were tested on their sensitivity to mispronunciations of the word forms of either early acquired words, or of recently segmented words (following a first experimental phase in which infants were presented with unfamiliar words embedded in sentential contexts). These studies revealed that a C-bias is present by 8 months when evaluated in a word form segmentation paradigm (Nishibayashi & Nazzi, 2016) and by 11 months when evaluated in a word form recognition paradigm (Poltock & Nazzi, 2015; Von Holzen & Nazzi, 2020). However, prior to that, a V-bias is found, at 6 months for word form segmentation (Nishibayashi & Nazzi, 2016), at 5 and 8 months for word form recognition (Bouchon et al., 2015; Von Holzen & Nazzi, 2020). This early V-bias is consistent with earlier studies on syllable discrimination in newborns having found higher sensitivity to vowels than consonants (Bertoncini et al., 1988). Taken together, the findings on French establish that the C-bias emerges in development, as attested by a reversal from a V- to a C-bias, and that this emergence takes place at a young age, around 8 to 11 months of age. Crucially, a very similar trajectory was found with Italian-

and Spanish-learning infants. In Italian, a study explored newborn's memorization of new word forms, and their sensitivity to a consonant or vowel mispronunciation of this word form (Benavides-Varela et al., 2012). The authors interpreted the pattern of brain responses measured with fNIRS (functional Near Infra-Red Spectroscopy) as a V-bias. This early V-bias is further corroborated by results from word learning experiments (based on anticipatory looking responses) showing that 6-month-olds give more weight to vowel information while 12-month-olds give more weight to consonant information, establishing a shift from a V- to a C-bias between these two ages in Italian (Hochmann et al., 2011, 2018). In Spanish, a study investigated whether infants are more sensitive to consonant or vowel mispronunciations of familiar word forms; it found a V-bias in 5-month-olds, no bias in 8½-month-olds and a C-bias in 12-month-olds (Bouchon et al., 2022). Taken together, the studies on French, Italian and Spanish establish that the C-bias emerges early in development in these three Romance languages, at some point in the second half of the first year of life.



6. Germanic languages

A second developmental trajectory was found for three Germanic languages, and would correspond to a late acquisition of a C-bias. In (British) English, studies exploring toddlers' ability to learn new words found a C-bias at 30 months of age (Nazzi et al., 2009), but equal sensitivity to consonant and vowel contrasts, hence no bias, at both 16 and 23 months (Flocchia et al., 2014). Studies on word recognition exploring toddler's sensitivity to mispronunciations found no bias at 12, 18 and 24 months, with some weak evidence of a C-bias at 15 months (Mani & Plunkett, 2007, 2010). Studies investigating the specificity of early word forms either failed to find evidence of specificity for the infant's own name at 5 months (Delle Luche et al., 2017), or found equal sensitivity to consonant and vowel mispronunciations when recognizing familiar word forms at 11 months (Ratnage et al., 2023). Taken together, these findings on British English establish the late emergence of a C-bias, between 24 and 30 months, following an equal sensitivity to consonants and vowels at younger ages.

Results from two other Germanic languages are consistent with the late emergence found in British English. In Danish, two studies explored this topic at 20 months of age. Studying Danish was motivated by the fact that it has more vowels than consonants, and is characterized by extensive consonant lenition. Both of these phenomena could disfavor a consonant bias. In a

first study in which the toddlers had to learn new words differing by either a consonant or a vowel, they succeeded only in the vowel-contrasted condition, attesting a V-bias (Højen & Nazzi, 2016). In a second study that was testing consonant and vowel specificity in word recognition, pitting consonant- and vowel-mispronunciations against each other, the toddlers failed to show a bias for either type of phoneme (Højen et al., 2023). In German, a study based on Højen et al. (2023) also failed to find a bias at 24 months of age (Piot et al., 2025). However, that study also tested French-learning 24-month-olds, who also failed to show a bias, when a C-bias was expected given the evidence presented above. Findings using this method should thus be interpreted with caution for the time being. That being said, it remains that no evidence of a C-bias could be found in Germanic languages before 30 months of age, contrary to the early emergence of the C-bias in Romance languages before the first birthday.



7. Sinitic languages

A third developmental trajectory was found for two Chinese languages, which are both tone languages, failing to find a C-bias in toddlerhood and childhood. The motivation for testing tone languages was that tonal information is mostly carried by the vowels, a phenomenon that could increase the relative processing weight given to vocalic information, and should thus reduce the C-bias. In Mandarin, studies on sensitivity to mispronunciations when recognizing familiar words provide some evidence of a V-bias in both Mandarin-monolingual and Mandarin-English-bilingual 24-month-olds (Wewalaarachchi et al., 2017). A similar study conducted on Mandarin-English-bilingual toddlers (aged 2.5–3.5 years) or preschoolers (aged 4–5 years) showed equal sensitivity to consonant and vowel mispronunciations (Singh et al., 2015). No bias was also found in both Mandarin-monolingual and Mandarin-English-bilingual 6-year-olds (Wewalaarachchi & Singh, 2020). In Cantonese, 30-month-olds tested in a word learning paradigm were found to learn pairs of words that differed by a vowel, but failed when the words differed by a consonant or a tone, establishing a V-bias (Chen et al., 2021). Taken together, the findings on these two Chinese languages suggest that a C-bias does not emerge in these languages, which is in line with the findings on adult speakers of these languages (Gomez et al., 2018; Poltrock et al., 2018; Wiener, 2019; Wiener & Turnbull, 2016).



8. Semitic languages

Last, work is starting on Semitic languages. A study investigating the sensitivity of Hebrew-learning 11-month-olds to consonant and vowel mispronunciations of familiar wordforms found no bias, infants being equally sensitive to consonant and vowel mispronunciations in stressed syllables and equally insensitive to consonant and vowel mispronunciations in unstressed syllables (Segal et al., 2020). This finding differs from the C-bias found in adult Hebrew speakers (Lador-Weizman & Deutsch, 2022, 2024). A study on Egyptian Arabic tested 21-month-old toddlers in a word learning conflict task where children are asked to pair objects whose labels share a consonant or a vowel (as done in Nazzi et al., 2009). Results show a clear consonant bias (Flocchia et al., 2025).

In summary, the developmental findings reported above establish that the C-bias is not present early in life, given that in the first days or months of life, a V-bias is found. This V-bias might result from the greater salience (longer duration and higher energy) of vowels compared to consonants. They further reveal an early emergence of a C-bias, during the first year of life, for 3 Romance languages (French, Italian and Spanish), a late emergence of a C-bias for English in the third year of life (with no bias found in the second year for Danish and German), no early bias in Hebrew, while data on Sinitic languages establish either a V-bias extending during the third year of life, or no bias even in childhood. These findings are in large part consistent with the adult findings, in which Indo-European languages show a C-bias while Sinitic languages show either a V-bias or no bias (note however that whether these linguistic families are the right level to distinguish the different trajectories needs to be further tested). The increased weight given to vowels in Sinitic languages was attributed in both the adult and developmental studies to the presence of lexical tones in these languages, which are mostly carried by the vowels. As previously discussed for the adult findings, if this increased weight of vowels extends to the majority of tone languages, then the C-bias might not be the predominant pattern in lexically-related processing.

At any rate, the crosslinguistic modulation in bias found in the developmental literature constitutes, at best, only partial evidence for the C-bias proposal as stated by Nespor et al. (2003), that is, for the strong claim that it is an in-built property of the language processing system. Rather, it supports the proposal that the processing biases that emerge with the acquisition of the native language are acquired. While the early acquisition

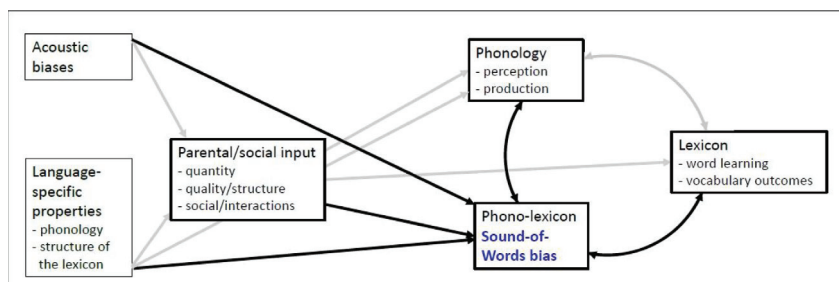


Fig. 2 The Sound-of-Words model, an integrative model of the acquisition of the phonolexicon (black arrows = direct links with Sound-of-Words bias; grey arrows = other likely indirect links).

of the C-bias in Romance languages was initially taken as evidence for the acoustic-phonetic over the lexical hypothesis, it has been argued more recently that both levels likely contribute to the acquisition of lexically-related processing biases (phono-lexical hypothesis; Poltrock & Nazzi, 2015). This is also supported by the late emergence found in English (and possibly other Germanic languages), since toddlers would have sufficiently large vocabularies to perform such analyses at the time the bias emerges. Moreover, when a C-bias was not found, it was attributed to phonological properties (consonant reduction in Danish; presence of lexical tones in Mandarin and Cantonese), and possibly to lexical factors (higher informativeness of vowels in distinguishing words in the lexicon of these languages).

In the future, further studies will need to extend the work already conducted to the exploration of even more numerous, and more varied, languages, and provide more detailed developmental trajectories than what is currently available, given that, for most languages, very few age points have been tested. Such studies will contribute to the determination of the relative prevalence of C- and V-biases, and to the specification of the phonological and lexical factors that guide and modulate their acquisition. In the following, we propose a new framework intended both to take into account what we currently know of the biases found in lexically-related processes during development, and to guide further research on this topic: the Sound-of-Words model.



9. The sound-of-words model

According to the Sound-of-Words model (see Fig. 2), the C-bias would only be one option in the binding of phonological and lexical abilities. Developmentally, infants would first give more weight to vowels

in language processing in general (due to their higher salience in the signal). The model then postulates how, for lexical processing, they would acquire a language-specific Sound-of-Words bias, be it a C- or a V-bias, as a result of parental/social speech input. Some language-general biases might influence that acquisition. For example, while the higher acoustic prominence of vowels over consonants might favor an initial V-bias, differences in the processing of consonants and vowels (more categorical for consonants, more continuous for vowels) might on the contrary favor the later emergence of a C-bias. Acoustic distance (such as differences in duration, intensity or spectral information) might also play a role in allowing discrimination and use of consonantal and vocalic contrasts to differentiate between words (see [Bouchon et al., 2015](#)).

Phonological and lexical properties of the language in acquisition would also impact the acquisition of the Sound-of-Words bias, which is reflected in the crosslinguistic variability in developmental trajectories revealed by the studies reported earlier. These properties still need to be better specified, and extending the repertoire of languages studied, varying their phonological properties, will allow this specification. Among the phonological factors that have been considered in the literature are the number of consonants and vowels in the inventory of the native language, and the ratio between them. The presence of lexical stress and vocalic reduction (in Germanic languages), consonantal reduction (in Danish), and lexical tones (in Sinitic tone languages) has been invoked. Syllabic structure (some languages allowing only consonant-vowel syllables, others allowing complex onset and coda consonants) might also modulate the bias. Among the lexical factors considered are cues that might signal the relative informativity of consonants and vowels to distinguish between words in the native lexicon, such as the number of consonantal and vocalic neighbors, or the number of unique consonant and vowel tiers. Their impact on the acquisition of the Sound-of-Words bias would be mediated by the quantity of input/words present in parental speech, but also by the quality of that input, in particular the variety/richness of the words spoken in the infants' environment. Their impact might also be related to the actual lexicon of each individual child. These effects of quantity and quality of input have been found to modulate other aspects of early language acquisition, such as word-form segmentation abilities and lexical acquisition (e.g., [Hurtado et al., 2008](#); [Weisleder & Fernald, 2013](#); [Newman et al., 2016](#); [Hoareau et al., 2019](#)).

The Sound-of-Words model further postulates that the emergence of the Sound-of-Words bias, at the interface of phonological and lexical

acquisition, would be linked to the concurrent development of related phonological and word learning abilities. It predicts that, at a given age, infants with better phoneme discrimination abilities and larger vocabularies would either acquire the Sound-of-Words bias earlier, or have a stronger Sound-of-Words bias. The model also states that the acquisition of the Sound-of-Words bias constitutes a major milestone in typical development corresponding to the binding between phonological and lexical acquisitions. And that this emergence, by allowing infants to focus on the sounds most relevant for lexical identity in their native language, would accelerate subsequent vocabulary development. Hence, in C-bias languages, infants with a stronger C-bias would be better word learners; similarly, in V-bias languages, infants with a stronger V-bias would be better word learners. This predicts that infants with a stronger bias at a given age would later have larger vocabularies than infants with a weaker bias. The age at which these predictive effects would be set into place likely depends on the timing of the acquisition trajectory for a specific language.

At present, very few studies have started to explore the links formulated by the Sound- of-Words model, and the evidence remains very scarce. For example, [Bouchon et al. \(2015\)](#) established that French-learning 5-month-olds' sensitivity to vowel-mispronunciations of their own name was stronger when the difference in spectral properties between their name and its mispronunciation was larger. However, this effect was not replicated in [Von Holzen and Nazzi \(2020\)](#). [Von Holzen and Nazzi \(2020\)](#) also investigated whether there was a link between infants' preference for their own name and the number of consonant or vowel tiers of the words they knew (based on parental vocabulary reports) at 5, 8 or 11 months. No links were found. With respect to the links between the C-bias and vocabulary development, [Poltrock and Nazzi \(2015\)](#) failed to find a link between the size of the C-bias in wordform recognition by French-learning 11-month-olds and their concurrent vocabulary size. Investigating the link between the C-bias and subsequent vocabulary size, no link was found in two studies, one investigating word-form segmentation at 8 months ([Von Holzen et al., 2018](#)), one investigating own name recognition at 5 and 8 months, with a reversed effect found at 11 months, infants with larger V-biases showing a faster vocabulary growth ([Von Holzen & Nazzi, 2020](#)). The study by [Von Holzen et al. \(2018\)](#) however established that higher sensitivity to all (consonant and vowel) mispronunciations of segmented word-forms was linked to faster vocabulary growth, which could support the proposal that the emergence of the C-bias is linked to overall better phoneme discrimination

abilities. Future studies will be needed, investigating these postulated links on larger cohorts of infants, tested on multiple abilities at several age points, and exploring how individual variability in Sound-of-Words bias relates to input, phonological and lexical abilities.

Future research will also have to specify how the acquisition of the Sound-of-Words bias relates to the acquisition of other aspects of the phonological properties of native words, such as that of prosodic properties starting around 6 months (e.g., Höhle et al., 2009; Jusczyk et al., 1993a; Skoruppa et al., 2009), or that of phonotactic properties starting around 9/10 months (e.g., Gonzalez-Gomez & Nazzi, 2012a; Jusczyk et al., 1994). Given the crucial bootstrapping role given to the acquisition of the C-bias in the division-of-labor proposal, the acquisition of the Sound-of-Words bias could be thought of as a prerequisite for other acquisitions, or at least as playing a more important role in further acquisitions. At present, only one series of studies explored this question, investigating whether the acquisition of phonotactic properties by French-learning infants is modulated by whether they relate to consonants or vowels (Gonzalez-Gomez & Nazzi, 2012a, 2016). These studies established that acquisition of a consonant-based property (the Labial-Coronal bias) emerges between 7 and 10 months, while acquisition of a vowel-based property (the Posterior-Anterior bias) emerges between 10 and 13 months, suggesting that having acquired a C-bias, French-learning infants learn consonant-related properties faster. Future studies will have to extend this line of research to more properties and more languages. In particular, it would be key to determine whether the opposite acquisition lag is found in V-bias languages (that is, earlier acquisition of vowel- over consonant-related phonotactic properties).

Given the importance of language-specific properties and linguistic input in determining the Sound-of-Word bias in the Sound-of-Words model, future studies will also need to address how this bias is acquired in infants who are learning several languages in parallel. Indeed, given the cross-linguistic variability in acquisition of the Sound-of-Words bias found in the literature reviewed above, some bilinguals will learn two languages with the same bias (for example, French and Italian), while others will learn two languages with opposite biases (for example, French and Cantonese). Moreover, even if they are learning two languages with the same bias, acquisition of the bias might not follow the same timing (for example, French and English). Hence, bilingualism or multilingualism might differently affect the acquisition of the Sound-of-Words(s) biases depending on the specific languages infants are

acquiring, and potentially on the language dominance pattern. The only current evidence on this issue comes from Mandarin-English bilinguals, tested only in Mandarin, showing a V-bias or most of the time no bias in lexical processing, as found for Mandarin-learning monolinguals (Singh et al., 2015; Wewalaarachchi et al., 2017; Wewalaarachchi & Singh, 2020).



10. Concluding remarks

In the present chapter, we reviewed evidence testing in both adults and infants the proposal that consonants play a larger role in lexically-related hypotheses than vowels. We presented evidence in support of this proposal, but also evidence of crosslinguistic variation suggesting that the C-bias might in fact only be one possible option. Accordingly, we proposed a new Sound-of-Words model of the acquisition of phonological biases in lexical processing that takes contribution from both nature and nurture, specifies links between the acquisition of the different linguistic levels and cascading developmental and predictive effects of early acquisitions on later developmental outcomes. This model could be seen as a guide to future research, and is likely to be later reformulated in light of the findings emerging from this future research. We would also want to encourage studying the acquisition of this bias in atypically developing populations (see Havy et al., 2010, on children with Williams syndrome; Havy et al., 2013, on deaf infants with cochlear implants; Gonzalez-Gomez & Nazzi, 2012b, on preterms). This would allow for the testing of the architecture of the Sound-of-Words model, but also for a better understanding of the specificity of language acquisition in neurologically-diverse populations, and for the setting up of efficient support interventions.

Acknowledgments

The author would like to thank Leonardo Piot and Caroline Floccia for comments on a previous draft of this article, Viviane Huet and Maxine Dos Santos for their invaluable contribution to the well functioning of the babylab, and coauthors, collaborators and colleagues who worked on the C-bias question along the years, parents for bringing their infants to the labs, and infants for accepting to participate and enlighten us by their responses to our experimental questions. The writing of this article was made possible thanks to a public grant overseen by the IdEx Université Paris Cité (ANR-18-IDEX-0001) as part of the Labex Empirical Foundations of Linguistics – EFL and an ANR grant (ANR-24-CE28-1770) to Thierry Nazzi.

References

- Alexeeva, S., Frolova, A., & Slioussar, N. (2017). Data from Russian help to determine in which languages the possible word constraint applies. *Journal of Psycholinguistic Research*, 46, 629–640.
- Benavides-Varela, S., Hochmann, J. R., Macagno, F., Nespors, M., & Mehler, J. (2012). New-born's brain activity signals the origin of word memories. *Proceedings of the National Academy of Sciences*, 109(44), 17908–17913. <https://doi.org/10.1073/pnas.1205413109>.
- Berdasco-Muñoz, E., Nishibayashi, L.-L., Baud, O., Biran, V., & Nazzi, T. (2018). Early segmentation abilities in preterm infants. *Infancy*, 23, 268–287.
- Bergelson, E., & Swingle, D. (2012). At 6–9 months, human infants know the meanings of many common nouns. In *Proceedings of the National Academy of Sciences of the USA*: 109 (pp. 3253–3258).
- Bertoncini, J., Bijeljac-Babic, R., Jusczyk, P. W., Kennedy, L. J., & Mehler, J. (1988). An investigation of young infants' perceptual representations of speech sounds. *Journal of Experimental Psychology*, 117, 21–33.
- Best, C. T., McRoberts, G. W., & Sithole, N. M. (1988). Examination of perceptual reorganization for nonnative speech contrasts: Zulu click discrimination by English-speaking adults and infants. *Journal of Experimental Psychology*, 14, 345–360.
- Bonatti, L. L., Peña, M., Nespors, M., & Mehler, J. (2005). Linguistic constraints on statistical computations: The role of consonants and vowels in continuous speech processing. *Psychological Science*, 16, 451–459.
- Bonatti, L. L., Peña, M., Nespors, M., & Mehler, J. (2007). On consonants, vowels, chickens, and eggs. *Psychological Science*, 18, 924–925.
- Bosch, L., Figueras, M., Teixido, M., & Ramon-Casas, M. (2013). Rapid gains in segmenting fluent speech when words match the rhythmic unit: Evidence from infants acquiring syllable-timed languages. *Frontiers in Psychology*, 4, 106.
- Bouchon, C., Floccia, C., Fux, T., Adda-Decker, M., & Nazzi, T. (2015). Call me Alix, not Elix: Vowels are more important than consonants in own-name recognition at 5 months. *Developmental Science*, 18(4), 587–598. <https://doi.org/10.1111/desc.12242>.
- Bouchon, C., Hochmann, J. R., & Toro, J. M. (2022). Spanish-learning infants switch from a vowel to a consonant bias during the first year of life. *Journal of Experimental Child Psychology*, 221, 105444. <https://doi.org/10.1016/j.jecp.2022.105444>.
- Caramazza, A., Chialant, D., Capasso, D., & Miceli, G. (2000). Separable processing of consonants and vowels. *Nature*, 403, 428–430.
- Chen, H., Lee, D. T., Luo, Z., Lai, R. Y., Cheung, H., & Nazzi, T. (2021). Variation in phonological bias: Bias for vowels, rather than consonants or tones in lexical processing by Cantonese-learning toddlers. *Cognition*, 213, 104486. <https://doi.org/10.1016/j.cognition.2020.104486>.
- Creel, S. C., Aslin, R. N., & Tanenhaus, M. K. (2006). Acquiring an artificial lexicon: Segment type and order information in early lexical entries. *Journal of Memory and Language*, 54, 1–19.
- Cutler, A., van Ooijen, B., & Norris, D. (1999). Vowels, consonants, and lexical activation. In *Proceedings of the Fourteenth International Congress of Phonetic Sciences*: 3 (pp. 2053–2056). San Francisco.
- Cutler, A., & Otake, T. (2002). Rhythmic categories in spoken-word recognition. *Journal of Memory and Language*, 46, 296–322.
- Cutler, A., Otake, T., & McQueen, J. M. (2009). Vowel devoicing and the perception of spoken Japanese words. *Journal of the Acoustical Society of America*, 125, 1693–1703.
- Cutler, A., Sebastian-Galles, N., Soler-Vilageliu, O., & van Ooijen, B. (2000). Constraints of vowels and consonants on lexical selection: cross-linguistic comparisons. *Memory and Cognition*, 28, 746–755. <https://doi.org/10.3758/BF03198409>.

- Delle Luche, C., Floccia, C., Granjon, L., & Nazzi, T. (2017). Infants' first words are not phonetically specified: Own name recognition in British English-learning 5-month-olds. *Infancy*, 22(3), 362–388. <https://doi.org/10.1111/inf.12151>.
- Delle Luche, C., Poltrock, S., Goslin, J., New, B., Floccia, C., & Nazzi, T. (2014). Differential processing of consonants and vowels in the auditory modality: A cross-linguistic study. *Journal of Memory and Language*, 72, 1–15. <https://doi.org/10.1016/j.jml.2013.12.001>.
- Dumay, N., Frauenfelder, U. H., & Content, A. (2002). The role of the syllable in lexical segmentation in French: Word-spotting data. *Brain and Language*, 81, 144–161.
- Eimas, P. D., Siqueland, E. R., Jusczyk, P., & Vigorito, J. (1971). Speech perception in infants. *Science*, 171, 303–306.
- El Aissati, A., McQueen, J. M., & Cutler, A. (2012). Finding words in a language that allows words without vowels. *Cognition*, 124, 79–84.
- Escudero, P., Mulak, K. E., & Vlach, H. A. (2016). Cross-situational learning of minimal word pairs. *Cognitive Science*, 40, 455–465.
- Feldman, N. H., Myers, E. B., White, K. S., Griffiths, T. L., & Morgan, J. L. (2013). Word-level information influences phonetic learning in adults and infants. *Cognition*, 127(3), 427–438.
- Floccia, C., Nazzi, T., Delle Luche, C., Poltrock, S., & Goslin, J. (2014). English-learning one-to-two-year-olds do not show a consonant bias in word learning. *Journal of Child Language*, 41(5), 1085–1114. <https://doi.org/10.1017/S0305000913000287>.
- Floccia, C., Khattab, G., Abdelwahab, A., & Nazzi, T. (in preparation, 2025). Evidence for a consonant bias in Arabic-learning toddlers.
- Fry, D. B., Abramson, A. S., Eimas, P. D., & Liberman, A. M. (1962). The identification and discrimination of synthetic vowels. *Language and Speech*, 5(4), 171–189. <https://doi.org/10.1177/002383096200500401>.
- Gomez, D. M., Mok, P., Ordin, M., Mehler, J., & Nespors, M. (2018). Statistical speech segmentation in tone languages: The role of lexical tones. *Language and Speech*, 61, 84–96.
- Gonzalez-Gomez, N., & Nazzi, T. (2012a). Acquisition of non-adjacent phonological regularities in the first year of life: Evidence from a perceptual equivalent of the labial–coronal effect. *Infancy*, 17, 498–524.
- Gonzalez-Gomez, N., & Nazzi, T. (2012b). Phonotactic acquisition in healthy preterm infants. *Developmental Science*, 15, 885–894.
- Gonzalez-Gomez, N., & Nazzi, T. (2016). Delayed acquisition of non-adjacent vocalic distributional regularities. *Journal of Child Language*, 43, 186–206.
- Goyet, L., Nishibayashi, L.-L., & Nazzi, T. (2013). Early syllabic segmentation of fluent speech by infants acquiring French. *PLoS ONE*, 8(11), e79646.
- Hallé, P., & Boysson-Bardies, B. de (1994). Emergence of an early lexicon: Infants' recognition of words. *Infant Behavior and Development*, 17, 119–129.
- Havy, M., & Nazzi, T. (2009). Better processing of consonantal over vocalic information in word learning at 16 months of age. *Infancy*, 14, 439–456.
- Hanulíková, A., McQueen, J. M., & Mitterer, H. (2011). Possible words and fixed stress in the segmentation of Slovak speech. *Quarterly Journal of Experimental Psychology*, 63, 555–579.
- Havy, M., Bertoncini, J., & Nazzi, T. (2011). Word learning and phonetic processing in preschool-age children. *Journal of Experimental Child Psychology*, 108(1), 25–43. <https://doi.org/10.1016/j.jecp.2010.08.002>.
- Havy, M., Moukawane, S., & Nazzi, T. (2010). Are 3-to-8 year old children with Williams syndrome good word-learners? *Neuroreport*, 21, 882–886.
- Havy, M., Nazzi, T., & Bertoncini, J. (2013). Phonetic processing during the acquisition of new words in 3-to-6-year-old French-speaking deaf children fitted with cochlear implant. *Journal of Communication Disorders*, 46, 181–192.
- Havy, M., Serres, J., & Nazzi, T. (2014). A consonant/vowel asymmetry in word-form processing: Evidence in childhood and in adulthood. *Language and Speech*, 57(2), 254–281. <https://doi.org/10.1177/0023830913507693>.

- Hoareau, M., Yeung, H. H., & Nazzi, T. (2019). Infants' statistical word segmentation in an artificial language is linked to both parental speech input and reported production abilities. *Developmental Science*, e12803. <https://doi.org/10.1111/desc.12803>.
- Hochmann, J. R., Benavides-Varela, S., Nespor, M., & Mehler, J. (2011). Consonants and vowels: Different roles in early language acquisition. *Developmental Science*, 14(6), 1445–1458. <https://doi.org/10.1111/j.1467-7687.2011.01089.x>.
- Hochmann, J. R., Benavides-Varela, S., Flo, A., Nespor, M., & Mehler, J. (2018). Bias for vocalic over consonantal information in 6-month-olds. *Infancy*, 23(1), 136–151. <https://doi.org/10.1111/infa.12203>.
- Höhle, B., Bijeljac-Babic, R., Herold, B., Weissenborn, J., & Nazzi, T. (2009). Language specific prosodic preferences during the first half year of life: Evidence from German and French infants. *Infant Behavior and Development*, 32, 262–274. <https://doi.org/10.1016/j.infbeh.2009.03.004>.
- Höhle, B., & Weissenborn, J. (2003). German-learning infants' ability to detect unstressed closed-class elements in continuous speech. *Developmental Science*, 6, 122–127.
- Hojen, A., Madsen, T., & Bleses, D. (2023). Danish 20-month-olds' recognition of familiar words with and without consonant and vowel mispronunciations. *Phonetica*, 80(5), 309–328. <https://doi.org/10.1515/phon-2023-2001>.
- Hojen, A., & Nazzi, T. (2016). Vowel bias in Danish word-learning: Processing biases are language-specific. *Developmental Science*, 19(1), 41–49. <https://doi.org/10.1111/desc.12286>.
- Hurtado, N., Marchman, V. A., & Fernald, A. (2008). Does input influence uptake? Links between maternal talk, processing speed and vocabulary size in Spanish-learning children. *Developmental Science*, 11, F31–F39.
- Jusczyk, P. W., & Aslin, R. N. (1995). Infants' detection of the sound patterns of words in fluent speech. *Cognitive Psychology*, 29(1), 1–23.
- Jusczyk, P. W., Cutler, A., & Redanz, N. (1993). Preference for the predominant stress patterns of English words. *Child Development*, 64, 675–687.
- Jusczyk, P. W., Luce, P. A., & Charles-Luce, J. (1994). Infants' sensibility to phonotactic patterns in the native language. *Journal of Memory and Language*, 33, 630–645.
- Kearns, R. K., Norris, D., & Cutler, A. (2002). Syllable processing in English. In *Proceedings of the Seventh International Conference on Spoken Language Processing* (pp. 1657–1660).
- Keidel, J. L., Jenison, R. L., Kluender, K. R., & Seidenberg, M. S. (2007). Does grammar constrain statistical learning? *Psychological Science*, 18(10), 922–923. <https://doi.org/10.1111/j.1467-9280.2007.02001.x>.
- Kooijman, V., Junge, C., Johnson, E. K., Hagoort, P., & Cutler, A. (2013). Predictive brain signals of linguistic development. *Frontiers in Psychology*, 4, 25.
- Kuhl, P. K., Williams, K. A., Lacerda, F., Stevens, K. N., & Lindblom, B. (1992). Linguistic experience alters phonetic perception in infants by 6 months of age. *Science*, 255, 606–608.
- Ladefoged, P., & Maddieson, I. (1996). *The Sounds of the World's Languages*. Oxford, UK: Blackwell Publishers.
- Lador-Weizman, Y., & Deutsch, A. (2022). The contribution of consonants and vowels to auditory word recognition is shaped by language-specific properties: Evidence from Hebrew. *Journal of Experimental Psychology: Human Perception and Performance*, 48(5), 401–426. <https://doi.org/10.1037/xhp0001000>.
- Lador-Weizman, Y., & Deutsch, A. (2024). The influence of language-specific properties on the role of consonants and vowels in a statistical learning task of an artificial language: A cross-linguistic comparison. *Quarterly Journal of Experimental Psychology*, 77(11), 2296–2311. <https://doi.org/10.1177/17470218241229721>.
- Liberman, A. M., Harris, K. S., Hoffman, H. S., & Griffith, B. C. (1957). The discrimination of speech sounds within and across phoneme boundaries. *Journal of Experimental Psychology*, 54(5), 358–368. <https://doi.org/10.1037/h0044417>.

- Mandel, D. R., Jusczyk, P. W., & Pisoni, D. B. (1995). Infants' recognition of the sound patterns of their own names. *Psychological Science*, 6, 314–317.
- Mani, N., & Plunkett, K. (2007). Phonological specificity of vowels and consonants in early lexical representations. *Journal of Memory and Language*, 57(2), 252–272. <https://doi.org/10.1016/j.jml.2007.03.005>.
- Mani, N., & Plunkett, K. (2010). Twelve-month-olds know their cups from their keps and tups. *Infancy*, 15(5), 445–470. <https://doi.org/10.1111/j.1532-7078.2009.00027.x>.
- Marks, E. A., Moates, D. R., Bond, Z. S., & Vazquez, L. (2002). Vowel mutability: The case of monolingual Spanish listeners and bilingual Spanish–English listeners. *Southwest Journal of Linguistics*, 21, 73–99.
- McQueen, J. M., & Cutler, A. (1998). Spotting (different kinds of) words in (different kinds of) context. In *Proceedings of the Fifth International Conference on Spoken Language Processing*: 6 (pp. 2791–2794). Sydney.
- McQueen, J. M., Otake, T., & Cutler, A. (2001). Rhythmic cues and possible-word constraints in Japanese speech segmentation. *Journal of Memory and Language*, 45, 103–132.
- Mehler, J., Peña, M., Nespor, M., & Bonatti, L. L. (2006). The “soul” of language does not use statistics: Reflections on vowels and consonants. *Cortex; A Journal Devoted to the Study of the Nervous System and Behavior*, 42, 846–854.
- Moates, D. R., Bond, Z. S., & Stockmal, V. (2002). Phoneme frequency in spoken word reconstruction. In C. Gussenhoven, & N. Warner (Eds.), *Laboratory Phonology VII* (pp. 141–169). Mouton.
- Morgan, J. L., & Demuth, K. (1996). Signal to syntax: Bootstrap–ping from speech to grammar in early acquisition. Mahwah, NJ: Erlbaum.
- Nazzi, T. (2005). Use of phonetic specificity during the acquisition of new words: differences between consonants and vowels. *Cognition*, 98, 13–30. <https://doi.org/10.1016/j.cognition.2004.10.005>.
- Nazzi, T., & Bertoncini, J. (2009). Phonetic specificity in early lexical acquisition: new evidence from consonants in coda positions. *Language & Speech*, 52, 463–480.
- Nazzi, T., & Cutler, A. (2019). How consonants and vowels shape spoken-language recognition. *Annual Review of Linguistics*, 1–23. <https://doi.org/10.1146/annurev-linguistics-011718-011919>.
- Nazzi, T., Floccia, C., Moquet, B., & Butler, J. (2009). Bias for consonantal information over vocalic information in 30-month-olds: Cross-linguistic evidence from French and English. *Journal of Experimental Child Psychology*, 102(4), 522–537. <https://doi.org/10.1016/j.jecp.2008.05.003>.
- Nazzi, T., & New, B. (2007). Beyond stop consonants: Consonantal specificity in early lexical acquisition. *Cognitive Development*, 22(2), 271–279. <https://doi.org/10.1016/j.cogdev.2006.10.007>.
- Nazzi, T., & Polka, L. (2018). The consonant bias in word learning is not determined by position within the word: Evidence from vowel-initial words. *Journal of Experimental Child Psychology*, 174, 103–111.
- Nazzi, T., Poltrock, S., & Von Holzen, K. (2016). The developmental origins of the consonant bias in lexical processing. *Current Directions in Psychological Science*, 25(4), 291–296. <https://doi.org/10.1177/0963721416655786>.
- Nespor, M., Peña, M., & Mehler, J. (2003). On the different roles of vowels and consonants in speech processing and language acquisition. *Lingue E Linguaggio*, 2, 203–230. <https://doi.org/10.1418/10879>.
- New, B., Araujo, V., & Nazzi, T. (2008). Differential processing of consonants and vowels in lexical access through reading. *Psychological Science*, 19(12), 1223–1227. <https://doi.org/10.1111/j.1467-9280.2008.02228.x>.

- New, B., & Nazzi, T. (2014). The time course of consonant and vowel processing during word recognition. *Language, Cognition and Neuroscience*, 29(2), 147–157. <https://doi.org/10.1080/01690965.2012.735678>.
- Newman, R. S., Rowe, M. L., & Ratner, N. B. (2016). Input and uptake at 7 months predicts toddler vocabulary: The role of child-directed speech and infant processing skills in language development. *Journal of Child Language*, 43(5), 1158–1173. <https://doi.org/10.1017/S0305000915000446>.
- Newman, R. S., Sawusch, J. R., & Wunnenberg, T. (2011). Cues and cue interactions in segmenting words in fluent speech. *Journal of Memory and Language*, 64, 460–476.
- Newport, E. L., & Aslin, R. N. (2004). Learning at a distance: I. Statistical learning of non-adjacent dependencies. *Cognitive Psychology*, 48, 127–162.
- Nishibayashi, L.-L., & Nazzi, T. (2016). Vowels, then consonants: Early bias switch in recognizing segmented word forms. *Cognition*, 155, 188–203. <https://doi.org/10.1016/j.cognition.2016.07.003>.
- Norris, D., McQueen, J. M., Cutler, A., & Butterfield, S. (1997). The possible-word constraint in the segmentation of continuous speech. *Cognitive Psychology*, 34, 191–243.
- Norris, D., McQueen, J. M., Cutler, A., Butterfield, S., & Kearns, R. (2001). Language-universal constraints on speech segmentation. *Language and Cognitive Processes*, 16, 637–660.
- Pace, A., Burchinal, P., Alper, R., Hirsh-Pasek, K., & Golinkoff, R. M. (2018). Measuring success: Within and cross-domain predictors of academic and social trajectories in elementary school. *Early Childhood Research Quarterly*, 46, 112–125.
- Piot, L., Nazzi, T., Boll-Avetisyan, N., van Ommen, S., & Schmandt, S. (2025). Phonological biases in lexical access in French and German: data from a familiar word-recognition conflict task. *Infant Behavior and Development*, 79, 102070.
- Poltrock, S., Chen, H., Kwok, C., Cheung, H., & Nazzi, T. (2018). Adult learning of novel words in a nonnative language: Consonants, vowels, and tones. *Frontiers in Psychology*, 9(JUL), 1–15. <https://doi.org/10.3389/fpsyg.2018.01211>.
- Poltrock, S., & Nazzi, T. (2015). Consonant/vowel asymmetry in early word form recognition. *Journal of Experimental Child Psychology*, 131, 135–148. <https://doi.org/10.1016/j.jecp.2014.11.011>.
- Pons, F., & Toro, J. M. (2010). Structural generalizations over consonants and vowels in 11-month-old infants. *Cognition*, 116, 361–367. <https://doi.org/10.1016/j.cognition.2010.05.013>.
- Ratnage, P., Nazzi, T., & Floccia, C. (2023). Vowels and consonants matter equally to British English-learning 11-month-olds' familiar word form recognition. *Journal of Child Language*, 1–24. <https://doi.org/10.1017/S0305000923000223>.
- Schmandt, S., Nazzi, T., & New, B. (2022). Consonant, vowel and lexical neighborhood processing during word recognition: New evidence using the sandwich priming technique. *Language, Cognition and Neuroscience*. <https://doi.org/10.1080/23273798.2022.2046115>.
- Segal, O., Keren-Portnoy, T., & Vihman, M. (2020). Robust effects of stress on early lexical representation. *Infancy*, 25(4), 500–521. <https://doi.org/10.1111/inf.12340>.
- Sharp, D. J., Scott, S. K., Cutler, A., & Wise, R. J. (2005). Lexical retrieval constrained by sound structure: The role of the left inferior frontal gyrus. *Brain and Language*, 92(3), 309–319. <https://doi.org/10.1016/j.bandl.2004.07.002>.
- Singh, L., Goh, H. H., & Wewalaarachchi, T. D. (2015). Spoken word recognition in early childhood: Comparative effects of vowel, consonant and lexical tone variation. *Cognition*, 142, 1–11. <https://doi.org/10.1016/j.cognition.2015.05.010>.
- Skoruppa, K., Pons, F., Christophe, A., Bosch, L., Dupoux, E., Sebastián-Gallés, N., Limisuri, R. A., & Peperkamp, S. (2009). Language-specific stress perception by 9-month-old French and Spanish infants. *Developmental Science*, 12, 914–919.

- Stager, C. L., & Werker, J. F. (1997). Infants listen for more phonetic detail in speech perception than in word-learning tasks. *Nature*, 388, 381–382.
- Swingle, D. (2005). 11-Month-olds' knowledge of how familiar words sound. *Developmental Science*, 8, 432–443.
- Tincoff, R., & Jusczyk, P. W. (1999). Some beginnings of word comprehension in 6-month-olds. *Psychological Science*, 10, 172–175. <https://doi.org/10.1111/1467-9280.00127>.
- Tincoff, R., & Jusczyk, P. W. (2012). Six-month-olds comprehend words that refer to parts of the body. *Infancy*, 17, 432–444. <https://doi.org/10.1111/j.1532-7078.2011.00084.x>.
- Toro, J. M., Nespor, M., Mehler, J., & Bonatti, L. (2008). Finding words and rules in a speech stream: Functional differences between vowels and consonants. *Psychological Science*, 19, 137–144. <https://doi.org/10.1111/j.1467-9280.2008.02059.x>.
- Trecca, F., McCauley, S. M., Andersen, S. R., Bleses, D., Basbøll, H., Højen, A., ... Christiansen, M. H. (2019). Segmentation of highly vocalic speech via statistical learning: Initial results from Danish, Norwegian and English. *Language Learning*, 69, 143–176.
- Van Ooijen, B. (1996). Vowel mutability and lexical selection in English: evidence from a word reconstruction task. *Mem. Cogn.*, 24, 573–583. <https://doi.org/10.3758/BF03201084>.
- Vihman, M., Nakai, S., dePaolis, R., & Hallé, P. (2004). The role of accentual pattern in early lexical representation. *Journal of Memory and Language*, 50, 336–353.
- Von Holzen, K., & Nazzi, T. (2020). Emergence of a consonant bias during the first year of life: New evidence from own-name recognition. *Infancy*, 25(3), 319–346. <https://doi.org/10.1111/inf.12331>.
- Von Holzen, K., Nishibayashi, L.-L., & Nazzi, T. (2018). Consonant and vowel processing in word form segmentation: An infant ERP study. *Brain Sciences*, 8, 24.
- Von Holzen, K., van Ommen, S., White, K. S., & Nazzi, T. (2023). The impact of phonological biases on mispronunciation sensitivity and novel accent adaptation. *Language Learning and Development*, 19(3), 303–322. <https://doi.org/10.1080/15475441.2022.2071717>.
- Weisleder, A., & Fernald, A. (2013). Talking to children matters: Early language experience strengthens processing and builds vocabulary. *Psychological Science*, 24(11), 2143–2152.
- Werker, J. F., & Tees, R. C. (1984). Cross-language speech perception: Evidence for perceptual reorganization during the first year of life. *Infant Behavior and Development*, 7, 49–63.
- Wewalaarachchi, T., & Singh, L. (2020). Vowel, consonant, and tone variation exert asymmetrical effects on spoken word recognition: Evidence from 6-year-old monolingual and bilingual learners of Mandarin. *Journal of Experimental Child Psychology*, 189 Article 104698. <https://doi.org/10.1016/j.jecp.2019.104698>.
- Wewalaarachchi, T. D., Wong, L. H., & Singh, L. (2017). Vowels, consonants, and lexical tones: Sensitivity to phonological variation in monolingual Mandarin and bilingual English-Mandarin toddlers. *Journal of Experimental Child Psychology*, 159, 16–33. <https://doi.org/10.1016/j.jecp.2017.01.009>.
- Wiener, S. (2020). Second language learners develop non-native lexical processing biases. *Bilingualism: Language and Cognition*, 23(1), 119–130. <https://doi.org/10.1017/S1366728918001165>.
- Wiener, S., & Turnbull, R. (2016). Constraints of tones, vowels and consonants on lexical selection in Mandarin Chinese. *Language and Speech*, 59(1), 59–82. <https://doi.org/10.1177/0023830915578000>.
- Yeung, H. H., & Nazzi, T. (2014). Infants can generalize phonetic contrasts learned from ostensive and referential labels. *Cognition*, 132, 151–163.
- Yeung, H. H., & Werker, J. F. (2009). Learning words' sounds before learning how words sound: 9-month-olds use distinct objects as cues to categorize speech information. *Cognition*, 113(2), 234–243. <https://doi.org/10.1016/j.cognition.2009.08.010>.

- Yip, M. C. W. (2004a). Possible-word constraints in Cantonese speech segmentation. *Journal of Psycholinguistic Research*, 33, 165–173.
- Yip, M. C. W. (2004b). Interference effects of possible-word constraints (PWC) in Cantonese speech segmentation. *Psychologia*, 47, 169–177.
- Zesiger, P., & Jöhr, J. (2011). Les représentations phonologiques des mots chez le jeune enfant. *Enfance*, 3(3), 293–309. <https://doi.org/10.4074/S001375451100303x>.